



OPEN Identifying the coupling mechanisms between land surface temperature and urban physical form in Chengdu's five central districts

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Urban thermal environment is a significant impact of human activities on the Earth's system. Understanding the relationship between thermal environments and urban physical form (UPF) at the block scale and proposing planning strategies is crucial. This study focuses on the five districts of Chengdu, using blocks as the analytical unit. It employs the radiation transfer equation method to invert land surface temperature (LST) and constructs a UPF index system from three aspects: land cover, building morphology, and road traffic. Using exploratory spatial data analysis and coupling coordination model, the study reveals the spatiotemporal evolution of LST and UPF and their dynamic-static coupling coordination relationship. The results show that the static evolution of LST and UPF exhibits spatiotemporal differences. LST showed a trend of "rise-then-fall", with its spatial distribution evolving from a "center-low, surroundings-high" to a "center-high, surroundings-low"; UPF showed a continuous upward trend, maintaining a "center-high, surroundings-low" pattern. The dynamic changes display a "central stability and peripheral fluctuation". The static coupling degree showed a general upward trend, primarily in the adaptation stage, while the dynamic coupling degree fluctuated significantly. The static coupling coordination degree presented an overall downward trend, with core blocks mainly in the adaptation phase and peripheral blocks in the antagonistic phase. The dynamic coupling coordination degree was generally lower than the static one, dominated by low-level coupling coordination, and characterized by "mild short-term fluctuations and pronounced long-term variability". The findings provide scientific evidence for climate-adaptive planning in Chengdu.

Keywords Land surface temperature, Urban physical form, Coupling coordination degree, Spatial autocorrelation, Block unit

Since the twenty-first century, urbanization has accelerated inevitably, driven by rapid socio-economic development and technological advancement. However, this rapid urbanization has also posed various environmental and social challenges, such as global climate warming¹, increasing ecosystem vulnerability² and the growing complexity of urban spatial structures³. Previous studies have highlighted a strong correlation and significant spatial heterogeneity between changes in urban physical form (UPF) and land surface temperature (LST). Neglecting the relationship between these factors can lead to spatial imbalances in heat risk management measures⁴, ultimately affecting the health⁵ and social well-being^{6,7} of urban residents. Therefore, identifying the dynamic impacts of urban spatial structure and morphological changes on the LST as well as the coupling and coordination, is of great significance. This issue not only represents a key research topic in climatology and urban planning⁸ but has also emerged as a focal area in urban geography⁹.

The urban thermal environment is one of the most prominent environmental challenges faced by human settlements in the context of rapid urbanization. Its formation mechanisms are comprehensively regulated by UPF, including building layout, green space configuration and road structure. Scholars both domestically and internationally have extensively studied the driving factors of the LST across multiple spatial scales and regional

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contexts^{10–12}. Kabisch's study demonstrated that tree cover and open green space in German cities have the lowest intensity of the urban heat island effect¹³. Wu explored how different street tree and grassland design schemes impact pedestrian thermal comfort in typical urban neighborhoods in Europe's temperate climate¹⁴. Previous studies have shown that land cover types and their biophysical properties significantly influence the LST¹⁵. For instance, vegetation cover^{16,17} and water body coverage¹⁸ are generally negatively correlated with LST, whereas impervious surface and building coverage are positively correlated with LST. In addition to these 2D factors, the influence of 3D urban spatial morphology on the thermal environment has also gained considerable attention. Studies have demonstrated that building morphological characteristics can affect the heat exchange process between the land surface and the atmosphere by regulating thermal radiation absorption and airflow efficiency^{19,20}. However, the findings of such studies are often constrained by scale effects²¹. Roy analyzed the impact of both 2D and 3D building morphology on LST in European cities, including Paris, Rotterdam, Milan, and Vienna²². Qian using meteorological data and UPF indicators revealed that the building coverage ratio significantly influences both daytime and nighttime urban heat island effects²³. However, the relationship between LST and its influencing factors is neither purely linear nor nonlinear. Most current studies primarily focus on correlation tests or the analysis of spatiotemporal pattern characteristics^{24,25}, yet there has been insufficient exploration of the dynamic evolution of the coupling relationship between the LST and UPF, as well as the synergistic mechanisms underlying their internal structure and developmental quality.

A common method for selecting spatial analysis units is to divide the study area into several equally sized regular grids for ease of remote sensing data processing²⁶. However, this approach overlooks the spatial heterogeneity of the built environment within the grids, which may lead to the fragmentation of building contours. Another approach involves dividing the urban area into units based on clusters of buildings with similar characteristics. This method better reflects the actual features of the built environment, although it leads to irregular grid structure. As the most basic administrative unit in China's urban structure, a block is typically enclosed by physical boundaries such as roads and rivers and its spatial extent is relatively stable. The land use and building morphology within a block often exhibit a high degree of homogeneity. Therefore, blocks are effective in reflecting the internal structural characteristics and functional homogeneity of the built environment. Moreover, block-level analysis can partially overcome the limitations of large-scale administrative zones and the lack of macro-level effects seen in fine-scale building unit analyses. As a result, the block has become an increasingly important unit in urban studies²⁷. However, research on the coupling and coordination relationship between the LST and UPF at the block scale remains limited, with practical urban planning strategies from this perspective largely unexplored.

This study focus on the five districts of Chengdu, obtained LST data retrieved from Landsat satellite imagery. A comprehensive indicator system of UPF was constructed from three components: land cover, building morphology and road traffic. At the block scale, the spatiotemporal patterns of LST and UPF were analyzed. The degree of coupling coordination model were applied to assess the coupling coordination levels and their dynamic evolution across different spatial units and temporal stages. This study offers valuable insight into the coupling relationship between urban LST and UPF changes and provides planning strategic for mitigating urban thermal environments and optimizing spatial structures.

Study area and data

Study area

The study area covers the central five districts of Chengdu in Sichuan Province, China (hereafter referred to as Chengdu), including Jinjiang, Qingyang, Chenghua, Wuhou and Jinniu districts, with a total area of 464.60 km². This region functions as the economic, cultural and political core of city, experiencing the most rapid urbanization and pronounced spatial morphology changes. Chengdu is located on the western edge of the Sichuan Basin and features a subtropical humid monsoon climate, with an average annual temperature of approximately 16 °C and an average annual precipitation of about 1300 mm. Chengdu is characterized by frequent calm wind conditions. From August 5 to 24, 2022, Chengdu experienced 20 consecutive days of extreme high temperatures, with seven of those days exceeding 40 °C—setting a historical record for the highest temperatures in nearly a century. Figure 1 illustrates the trends in Chengdu's annual average temperature, decadal average temperature and monthly average maximum temperature from 1970 to 2023. The data show a significant upward trend in annual average temperature, rising from 14.8 °C in 1970 to 16.5 °C in 2023—an increase of 1.7 °C, or approximately 0.23 °C per decade. Likewise, the extreme values of monthly average maximum temperatures have also increased, rising by roughly 0.25 °C per decade, peaking at 28.32 °C in 2022. The persistent high-temperature conditions in Chengdu's not only pose direct threats to public health and quality of life but also intensify energy consumption, power grid overloads and disrupt industrial production and urban function.

From 2010 to 2023, the permanent population in the five central districts of Chengdu increased by 1.47 million and the regional gross domestic product grew by 76.95 billion yuan, with an urbanization rate reaching 100%. The growth in population and economy intensified the demand for construction land, leading to continuous urban expansion, which further contributed to the sustained rise of LST²⁷. Therefore, this study focus on the five central districts of Chengdu—where thermal environmental are particularly severe—as the study area (Fig. 2), with the aim of exploring the spatiotemporal evolution and coupling coordination relationship of the LST and UPF.

Data and source

The data used in this study comprise Landsat satellite remote sensing images with a spatial resolution of 30 m downloaded from the United States Geological Survey (USGS). To ensure image quality and maintain consistency in acquisition dates, three cloud-free images with less than 10% cloud cover were selected for LST retrieval. These images were taken on April 2, 2015, July 28, 2020, and April 13, 2025 (with overpass time at 11:33

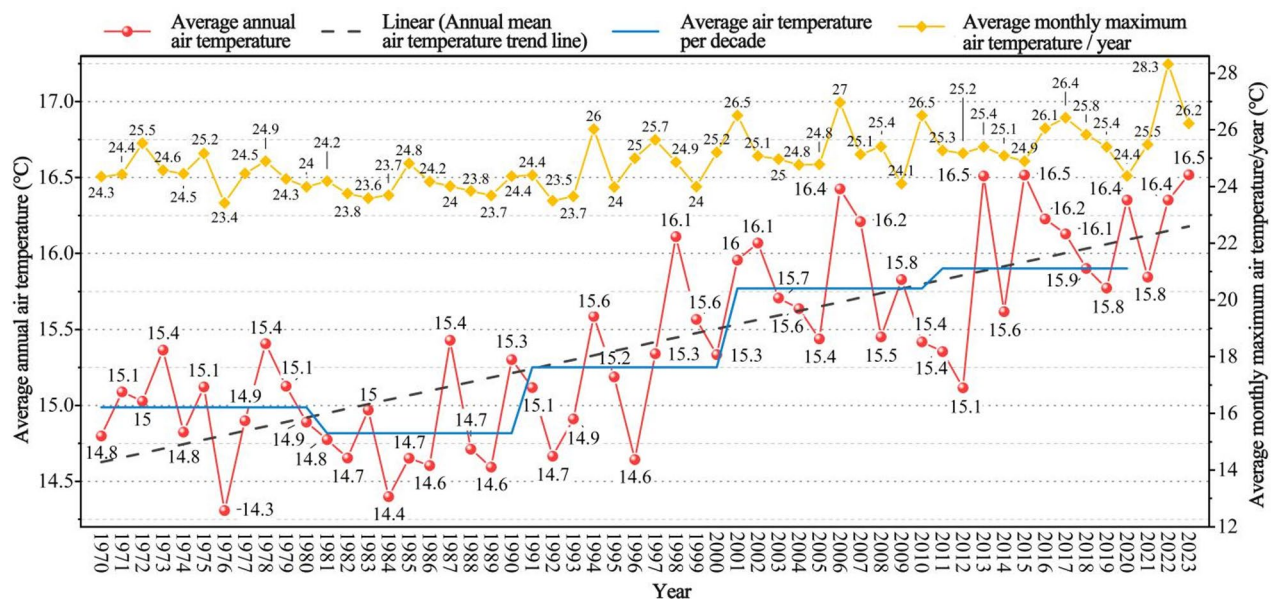


Fig. 1. Trend analysis of air temperature changes in Chengdu from 1970 to 2023.

a.m. Beijing time). Land cover data with a 30 m resolution provided by Global Land Cover project²⁸ and covering the period from 1990 to 2022 was used to calculate the areal proportions of different land use types. Since the dataset only extends through 2022, the 2022 data were used to represent 2025, ensuring that spatial features and land surface information align as closely as possible with the analysis year. Building data were obtained from Amap, including information on building heights and footprints area, which were used to calculate building morphological indicators. Road data were sourced from OpenStreetMap to delineate block units and calculate road density. The meteorological observations data were obtained from the basic meteorological element dataset of national ground meteorological stations in China²⁹. Monthly temperature data for Chengdu from 1970 to 2023 were selected to analyze long-term urban climate change trends. Additionally, average temperature data corresponding to the same dates as the remote sensing imagery were used to validate the effectiveness of the LST. Since there are no basic meteorological stations within the five central districts of Chengdu, air temperature data from adjacent districts (Pidu, Wenjiang, Xindu, Shuangliu, and Longquanyi) were utilized for comparison. The specific data types and related information are listed in Table 1.

Methods

Determination of spatial analysis units

The block scale is more closely aligned with the practical operations of urban planning, urban renewal and refined management³⁰. Typically, urban block ranges from 150 to 200 m, although some areas may feature slightly larger blocks³¹. In this study, the extracted blocks generally ranged from 300 to 500 m, with densely built-up areas usually around 200 m. For more sparsely built-up areas on the urban periphery, the block size were extended to about 1 km. The specific process for defining blocks was as follows: First, highway, main road, secondary road, tertiary road, and residential road data were selected from the OSM dataset. Then, based on the concepts of dilation and erosion in mathematical morphology, the centerlines of multi-line roads were extracted. Following this, a topological check was conducted on the road network and the road lines were converted into block polygons. Finally, for blocks with noticeable deviations or poor segmentation accuracy, manual corrections were made using Gaode online maps. A total of 5135 blocks were defined in the study area, with an average area of approximately 0.09 km².

LST retrieval

LST retrieval employs the radiative transfer method, which is based on the principles of thermal radiation and energy conservation. This method is characterized by a solid physical foundation and high retrieval accuracy³². The core idea is to calculate LST based on the intensity of thermal infrared radiation detected by a thermal infrared radiometer. The retrieval process accounts for three components: upward atmospheric radiation, downward atmospheric radiation, and the radiation reaching the radiometer after surface radiation has transmitted through the atmosphere. The satellite's thermal infrared spectral radiance is given by the following Eq. (1):

$$L_{\lambda} = [\varepsilon \cdot B(T_s) + (1 - \varepsilon) \cdot L_{\downarrow}] \tau + L \quad (1)$$

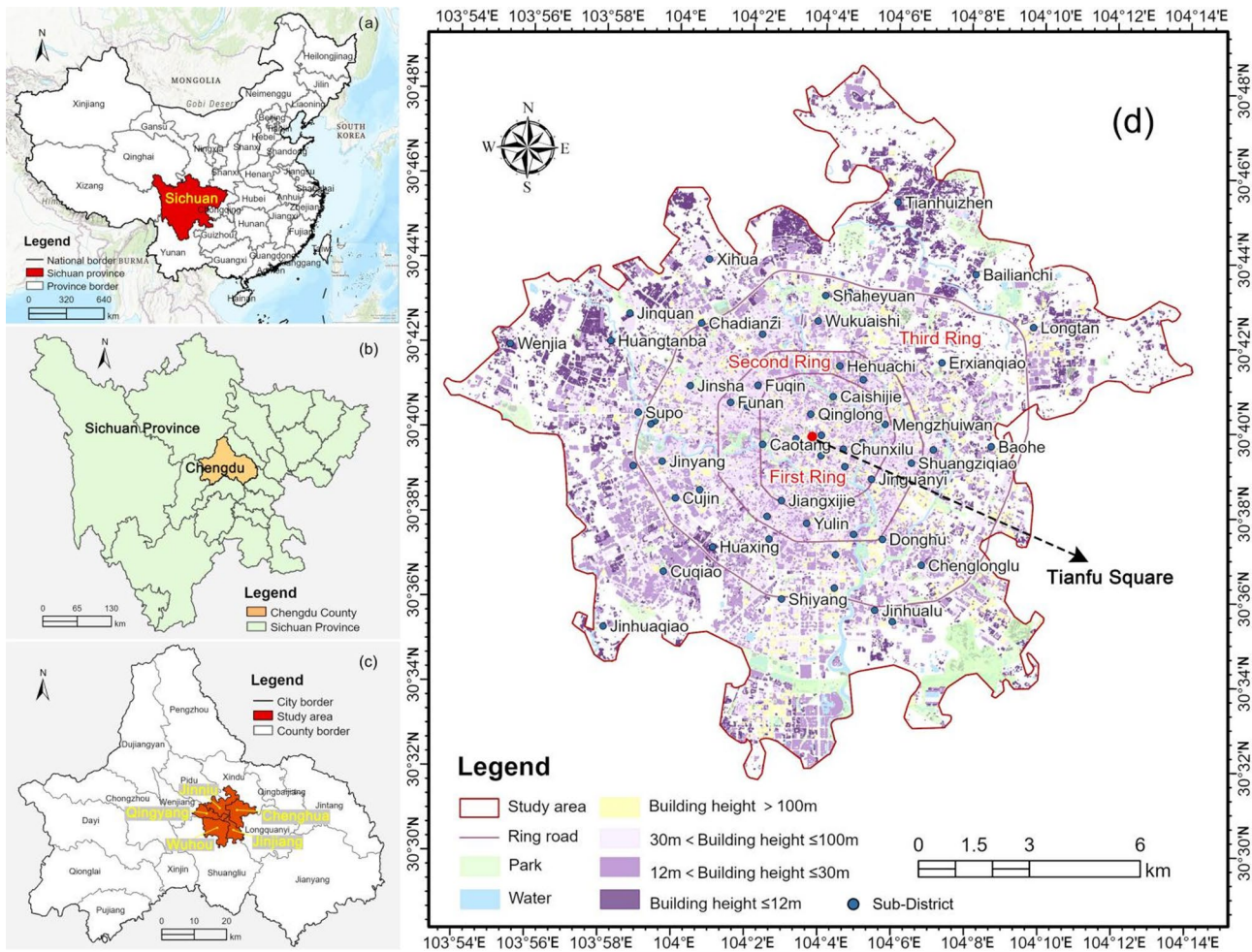


Fig. 2. Location map of the study area: (a) Location of Sichuan Province in China; (b) Location of Chengdu City within Sichuan Province; (c) Location of the five central districts within Chengdu City; (d) Distribution of building heights in the five central districts of Chengdu City. This map is based on the standard map with the review number GS (2024) 0650 downloaded from the Standard Map Service website of the National Bureau of Surveying and Mapping Geographic Information. The base map has not been modified.

Type of data	Source	Date/Year
Monthly meteorological data	National Ground Meteorological Stations in China http://data.tdpc.ac.cn	1970–2023
Remote sensing images	United States Geological Survey https://earthexplorer.usgs.gov/	2015.04.02 Landsat 7(ETM)
		2020.07.28 Landsat 8(OLI)
		2025.04.13 Landsat 8(OLI)
Land use data	http://www.globallandcover.com/	2015, 2020 and 2022
Building data (footprint area, heights)	Amap	2015, 2020 and 2025
Road data	OpenStreetMap	2015, 2020 and 2025

Table 1. Details of all data.

where T_s is the LST in degrees Celsius, ϵ the emissivity of the surface, $B(T_s)$ the radiance of a blackbody with the temperature T_s , τ the atmospheric transmittance, The unit of the upward atmospheric radiance ($L_{\uparrow}$), and the downward atmospheric radiance ($L_{\downarrow}$) is $W \cdot m^{-2} \cdot sr^{-1}$.

Solving Eq. (2) yields $B(T_s)$. Thereafter, LST (T_s) can be calculated using Eq. (3)

$$B(T_s) = [L_{\lambda} - L_{\uparrow} - \tau \cdot (1 - \epsilon) \cdot L_{\downarrow}] / \tau \epsilon \tag{2}$$

$$T_s = \frac{K_2}{\ln \left(\frac{K_1}{B(T_s)+1} \right)} \tag{3}$$

where (K_1K_2) corresponds to the sensor calibration constants applicable to the 6th band and the 10th band of Landsat 7 and 8 images. According to the official documentation from USGS, the values for three parameters—upward atmospheric radiance ($L_{\uparrow}$), the downward atmospheric radiance ($L_{\downarrow}$) and atmospheric transmittance—can be obtained from the mean values of the corresponding raster data with suffixes URAD.TIF, DRAD.TIF and ATRAN.TIF in the Landsat Collection 2 Level-2 product. The mean values of the first two parameters should be reduced by a factor of 1,000 and the mean value of the third parameter should be reduced by a factor of 10,000. The surface emissivity (ϵ) was calculated using the NDVI method as provided in a previous study³³. Some studies have performed LST retrieval using nighttime remote sensing³⁴ and nighttime light intensity imagery³⁵. The major challenge of this method lies in distinguishing between optical and thermal anomalies; therefore, our study focuses primarily on daytime LST. After retrieving the LST for Chengdu, we used the zonal statistics tool in ArcGIS to calculate the mean LST for each block unit within the study area. The results were subsequently visualized to present the spatial distribution of the LST across different blocks.

Construction of a comprehensive UPF indicator system

To systematically evaluate the impact of UPF on the LST, this study develops a comprehensive UPF indicator system based on 2D and 3D urban morphology characteristics, encompassing three components: land cover, building morphology and road traffic (Table 2). At the land cover level, different types of surface materials have significant effects on local LST. For example, impervious surfaces characterized by high thermal radiation absorption and low albedo often exacerbate the LST³⁶. Conversely, vegetation³⁷ and water bodies³⁸ can effectively reduce surrounding LST through transpiration and evaporative cooling. Therefore, three indicators—the proportion of impervious surface area, forest and grassland area and water body area—were selected to represent the thermal regulation functions of different land cover types. Buildings, as fundamental elements of urban spatial structure, directly influence the process of heat absorption, storage and dissipation. Building height reflects variations in vertical scale, while building density indicates the degree of horizontal aggregation; together, they determine the accumulation and diffusion capacity of heat within urban space. Additionally, the sky view factor (SVF) assess the density of 3D urban morphology and the extent of sky obstruction, quantifying the degree to which building block sky visibility, thereby influencing solar radiation access and heat dissipation. Road traffic, a major form of artificial hard surface in cities, significantly influences the thermal environment through its spatial pattern. Road density not only reflects urban accessibility and development intensity but also represents underlying physical properties such as surface albedo and heat capacity³⁹. In addition to the various road categories used for block unit delineation, we also incorporated various types of minor roads, including service roads, sidewalks and bicycle lanes from the SOM dataset to calculate the road density for each block unit.

To quantify the relative importance of each factor, this study employed the analytic hierarchy process, using expert scoring and the construction of a judgment matrix to assess the importance of different indicators in the influence of UPF and to determine their respective weights. To ensure the consistency and comparability of the data, the Min-Max Normalization method was applied to scale the values of the independent variables to a range of 0–1, thus eliminating dimensional differences (Eq. 4). Finally, by integrating the weighted values, the comprehensive UPF index for Chengdu was calculated. To reveal the spatial differences of UPF, the zonal statistics tool in ArcGIS was used to calculate the mean UPF for each block unit within the study area.

$$X_{\text{norm}} = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}} \tag{4}$$

In the equation, X represents the original indicator value, X_{min} and X_{max} represent the minimum and maximum values of the independent variable, respectively. Since the calculation of the comprehensive UPF index relies on the weights of each independent variable and changes in these weights may affect the coupling results. To assess the sensitivity of the results, a sensitivity analysis was conducted by adjusting the weights of the indicators and observing their impact on the comprehensive UPF and its coupling relationship with LST. Specifically, the weight of water bodies was set to 0.15 and the weight of road density was set to 0.12. The results indicated that while adjusting the weight of individual indicators caused slight fluctuations in the UPF, the long-term trend of the coupling relationship between UPF and LST remained largely unchanged.

Primary indicator	Secondary indicator	Tertiary indicator	Weight
Urban physical form	Land Cover	Percentage of impervious surface (%)	0.16
		Percentage of water bodies (%)	0.11
		Percentage of forest and grassland (%)	0.12
	Building Morphology	Average building height (m)	0.17
		Building density (%)	0.17
		Sky view factor	0.11
	Road traffic	Road density (km km ⁻²)	0.16

Table 2. UPF factors and their corresponding weights.

Exploratory spatial data analysis

Exploratory Spatial Data Analysis (ESDA) is a method used to analyze spatial relationships among attribute data, highlighting spatial interactions through numerical correlation analysis and revealing spatial association patterns⁴⁰. In this study, ESDA was employed, focusing on global and local spatial autocorrelation analyses to explore the spatial autocorrelation of LST, UPF and their respective rates of change. First, Global Moran's I was used to assess the spatial autocorrelation of these variables in the study area, indicating their average degree of spatial clustering. A Global Moran's I value greater than 0 indicates positive spatial autocorrelation, a value equal to 0 indicates no spatial correlation and a value less than 0 signifies negative spatial autocorrelation. Next, the Getis-Ord G_i^* index was used for hotspot analysis to identify the clustering of local spatial data and analyze the spatial instability. A significantly positive Getis-Ord G_i^* index that the spatial location and its surrounding values are high, signifying a hotspot area. Conversely, a significantly negative value indicates a cold spot area.

Coupling degree and coupling coordination degree model

"Coupling" refers to the interaction between two or more systems or elements, reflecting the dynamic relationships and mutual coordination processes among internal system components⁴¹. The coupling degree quantifies the strength of intersystem interaction, with its magnitude indicating the evolutionary shift from disorder to order within the system. A higher coupling degree suggests strong internal interactions, promoting stability and synergistic development, while a lower degree signifies insufficient interaction and unbalanced development. The coupling coordination degree, on the other hand, represents the level of harmonious coupling within the interaction process, reflecting the quality of coordination⁴². Based on coupling degree models from physics⁴³, this study defines the coupling relationship between LST and the comprehensive UPF as follows:

(1) Static coupling degree model

$$C1 = \sqrt[2]{\frac{P_i L_i}{[(P_i + L_i)/2]^2}} \quad (5)$$

In the equation, $C1$ represents the static coupling degree between LST and UPF, $C1 \in [0, 1]$; P_i, L_i denote the scores of LST and UPF within each block unit of the study area, respectively.

(2) Dynamic coupling degree model.

Given the dynamic nature of the LST and physical forms, observations should account not only for the initial and final states but also for the magnitude and direction of changes occurring over the observation period. Therefore, this study focus on two specific time periods, "2015–2020" and "2020–2025" for dynamic coupling analysis. These phases represent key stages in urban development: the first reveals the impact of urban expansion on the LST, while the second reflects the optimization of the LST during the urban stock development process. Following the short-term analysis, a dynamic coupling analysis for the long-term period (2015–2025) was conducted to explore the evolutionary trends and overall patterns of the LST and UPF. Based on the changes in LST and UPF, a dynamic coupling level model is developed as follows:

$$C2 = \sqrt{\frac{|\Delta P_i| \times |\Delta L_i|}{(|\Delta P_i| + |\Delta L_i|)^2}} \quad (6)$$

$$\Delta P_i = \frac{P_i^{t_2} - P_i^{t_1}}{P_i^{t_1}}; \Delta L_i = \frac{L_i^{t_2} - L_i^{t_1}}{L_i^{t_1}} \quad (7)$$

In the equation, $C2$ represents the dynamic coupling degree between LST and UPF, $C2 \in [0, 1]$; t_1, t_2 denote the start and end time points of the study period, respectively; $\Delta P_i, \Delta L_i$ indicate the rates of change in LST and UPF scores within the i block unit, respectively; $|\Delta P_i|, |\Delta L_i|$ denotes the absolute value of the corresponding terms.

(3) Coupling coordination degree model.

The coupling coordination degree model is further defined to quantify the coordination relationship between LST and UPF over long-term evolution. The calculation methods for static coupling coordination degree and dynamic coupling coordination degree are given by the following Eq. (8) and Eq. (9), respectively:

$$D1 = \sqrt{C1 \times T1} \quad (8)$$

$$D2 = \sqrt{C2 \times T2} \quad (9)$$

$$T1 = \alpha X1 + \beta Y1, T2 = \alpha X2 + \beta Y2 \quad (10)$$

In the equation, $D1$ represent the static coupling coordination degree, $D2$ represent the dynamic coupling coordination degree. $T1$ is the comprehensive evaluation indices for static LST and UPF, $T1$ is the comprehensive evaluation indices for and dynamic LST and UPF. $X1$ and $Y1$ represent the mean static value of LST and UPF within each block unit, while $X1$ and $Y1$ represent the mean dynamic values LST and UPF within each block unit. α, β are undetermined coefficients, both set to 0.5 in this study. Referring to the findings of Li et al.⁴⁴

and the coupling degree levels observed in Chengdu, the coupling degree and coupling coordination degree are categorized into four stages: low-level coupling stage(0.0, 0.15], antagonistic stage(0.15, 0.3], adaptation stage(0.30, 0.45], and high-level coupling stage(0.45, 0.6]. Due to the relatively high values of dynamic coupling degree, this study classifies them separately into the four stages: low-level coupling stage(0.0, 0.30], antagonistic stage(0.30, 0.50], adaptation stage(0.50, 0.80], and high-level coupling stage(0.80, 1.00].

Result and analysis

Spatiotemporal evolution characteristics of urban LST

Figure 3 show the LST retrieval results at the block level in Chengdu for the years 2015, 2020 and 2025 along with the spatial distribution of their respective change rates. The mean LST values for each years were 35.03 °C, 41.59 °C and 33.80 °C respectively. The corresponding average temperatures for the same days were 28.82 °C, 36.62 °C and 27.55 °C. These differences fall within a normal range, ensuring the feasible of the subsequent analysis. Overall, the LST exhibits an initial increase followed by a decline, which is consistent with findings reported in other literature⁴⁵ and mirrors the LST variation trend observed in Wuhan over the past 35 years⁴⁶. The peak observed in 2020 was primarily attributed to prolonged extreme heat events that year, while the decrease observed in 2025 can be linked to urban planning measures, such as the expansion of green space and the construction of ecological corridors. Spatially, the LST in 2015 and 2020 exhibited a “low center, high periphery” pattern, with elevated temperature primarily concentrated in blocks characterized by high construction intensity. In 2015, typical high-temperature blocks included the Jinhuajiao and Cuqiao blocks in Wuhou district, as well as the Erxianqiao and Baohe block in Chenghua district. The blocks in Qingyang district showed moderate mean LST. Low-temperature blocks were scarce and primarily restricted to large natural water bodies and a sizeable low-temperature area in the Huaxing block in Wuhou district. As urban expansion continued outward, a large number of high and sub-high temperature blocks were newly added to the urban periphery in 2020. This trend is consistent with the effects of urbanization on warming observed in large-scale studies⁴⁷. Simultaneously, the number of low-temperature blocks increased, with major rivers such as the Fuhe, Shahe, Jinjiang, Nanhe and Dongfeng Canal forming the backbone of Chengdu’s cold island network and playing a crucial role in urban cooling. By 2025, the overall urban LST had been moderately alleviated. The LST in near-suburban zones decreased significantly and the central urban area was dominated by medium and sub-medium temperature

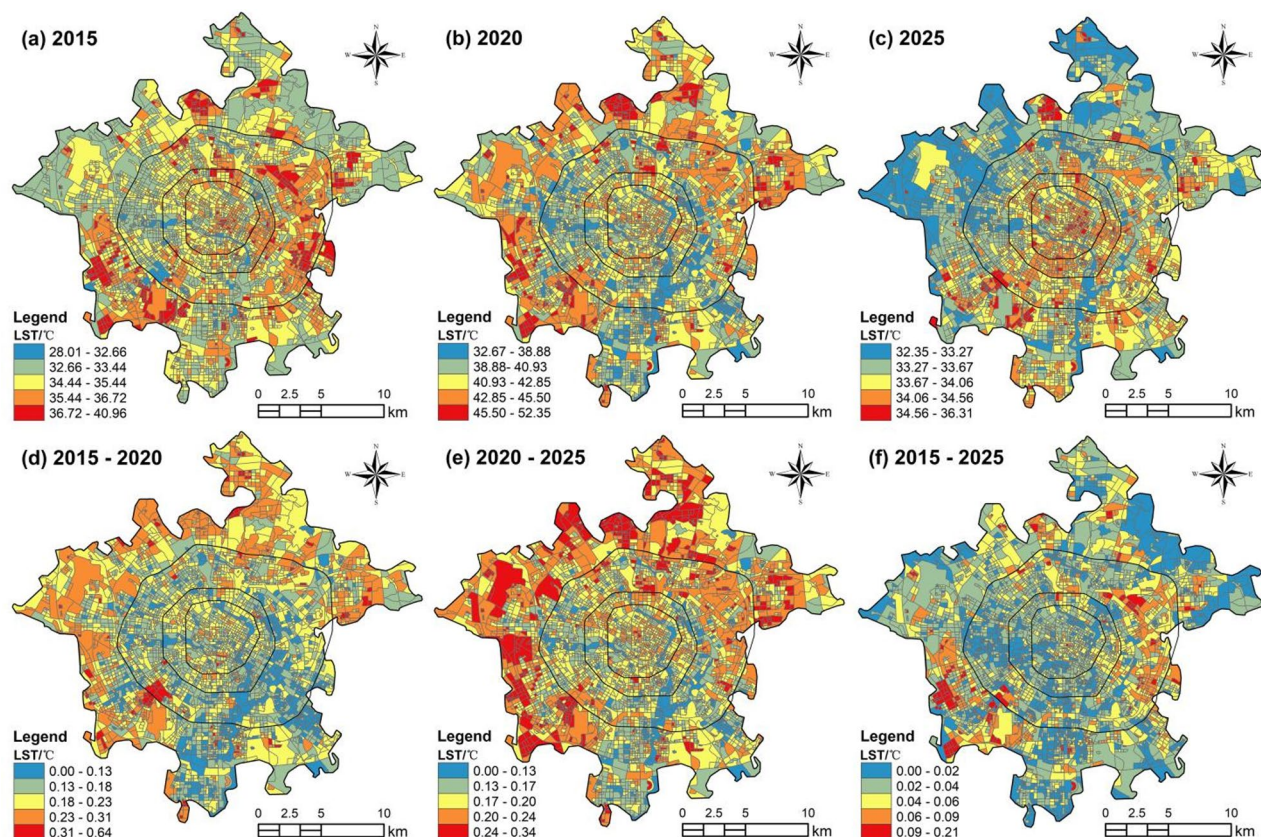


Fig. 3. Spatiotemporal patterns of LST and its rate of change in Chengdu: (a–c) Represent the spatial distribution of static LST for the years 2015, 2020 and 2025, respectively; (d–f) show the spatial distribution of LST change rates for the periods 2015–2020, 2020–2025 and 2015–2025, respectively. (Shades from red to blue represent high temperature zones, sub-high temperature zones, medium temperature zones, sub-medium temperature zones and low temperature zones).

zones with scattered patches of high-temperature zones. The spatial pattern shifted from the original “low center, high periphery” to a “high center, low periphery” distribution.

Regarding the rate of change in LST, the short-term periods from 2015 to 2020 and from 2020 to 2025 exhibited relatively high rates of change, at 0.1849 and 0.1835, respectively. In contrast, the long-term period from 2015 to 2025 showed a much smaller rate of 0.0372, primarily influenced by extreme climate events in 2020. Spatially, the short-term period exhibited a “stable center, fluctuating periphery” pattern in LST changes. The central urban zones experienced relatively minor changes, maintaining overall stability, while the peripheral zones, especially in the northwest, showed more significant fluctuations in LST. Over the long-term period from 2015 to 2025, LST changes were relatively moderate, with notable changes occurring only in certain peripheral blocks. The overall spatial pattern formed a concentric ring of “low–high–low” radiating outward from the city center.

The results of the global spatial autocorrelation test for urban LST and its change rates are summarized in Table 3. The findings indicate that, whether at individual time points or dynamic change rates, Moran’s *I* values are positive, with Z-scores significantly exceeding 2.58, all passing the significance test at $P < 0.01$. This demonstrates that the spatial distribution of LST and its change rates in Chengdu’s blocks exhibit significant positive spatial autocorrelation, meaning that blocks with relatively high or low LST values and change rates tend to be spatially adjacent and clustered⁴⁸. Temporally, Moran’s *I* values for LST showed a declining trend from 2015 to 2020, indicating a reduction in spatial clustering of LST during this period. Conversely, from 2020 to 2025, Moran’s *I* values increased, indicating enhanced spatial clustering. Regarding the LST change rates, the highest Moran’s *I* value was observed during the 2020–2025 period, indicating the most pronounced spatial clustering of LST change rates during this time. In contrast, the lowest Moran’s *I* value occurred over the 2015–2025 period, indicating the weakest spatial clustering, while the 2015–2020 period demonstrated a moderate level of spatial clustering.

By analyzing the spatial distribution characteristics of hot and cold spots of LST and its change (Fig. 4), the spatiotemporal patterns of LST and its variations across urban blocks can be further revealed from a local perspective. At a single time point, the hotspot blocks of LST in 2015 and 2020 were dispersed in patch-like patterns around the urban periphery. This distribution is attributed to the varying land use and its functions during different periods. The urban periphery typically features more mixed-use areas, which leads to a “fragmented” hotspot pattern. By 2025, the hotspots formed a northeast-southwest oriented belt concentrated in the city center, with a continuous hotspot area in the southern part of the Cuqiao block in Wuhou district. This reflects the trend of urban development gradually concentrating toward the central area. The distribution of cold spot blocks became more continuous, particularly in 2025, with extensive and patchy distributed across various blocks in Qingyang district. This indicates that with urban development and planning, low-temperature blocks are forming a stable and widespread distribution in the urban periphery. Regarding the spatial distribution of hot and cold spots in LST change rates, the patterns observed during the short-term periods (2015–2020 and 2020–2025) were similar. Hot spots were primarily concentrated along the northwestern urban edge, while cold spots were mainly found in the southeast sector. In both period, hot and cold spot blocks showed spatially contiguous distributions. However, in the long-term period (2015–2025), the spatial pattern of hot and cold spots was more fragmented, with only one extensive cold spot zone and two sizable hotspots zones identified.

Characteristics of changes in UPF
Characteristics of changes in different UPF factors

- (1) Land cover factors.

Figure 5 presents the spatiotemporal variation patterns of different land cover factors across blocks in Chengdu for the years 2015, 2020 and 2025. In terms of impervious surface area, the blocks exhibited relatively high coverage levels across the three years, with average proportions of 0.94%, 0.92% and 0.75% respectively, showing a general downward trend over time. Although previous studies have indicated an overall increase in impervious surfaces due to urban expansion in Chengdu⁴⁹, this study’s block-scale analysis reveals a declining trend in average impervious surface coverage. This discrepancy may be stem from differences in the scale of analysis: earlier research often relied on aggregated data at the city or administrative district level, emphasizing urban expansion and land use change. In contrast, the block scale analysis better captures localized changes such as urban renewal, green space expansion and redevelopment of underutilized lands. Further analysis indicates that blocks with decreasing impervious surface proportions are primarily mixed-use blocks, reflecting targeted urban renewal strategies. The maximum impervious surface proportions across blocks were 17.97%, 15.89%

	Year	Global Moran’s I	Z value	P value
Land surface temperature	2015	0.367 1	99.691 4	0.000 0
	2020	0.366 8	99.597 6	0.000 0
	2025	0.468 5	127.194 9	0.000 0
	2015–2020	0.316 2	85.872 5	0.000 0
	2020–2025	0.343 8	93.362 9	0.000 0
	2015–2025	0.303 4	82.417 9	0.000 0

Table 3. Analysis results of global spatial autocorrelation.

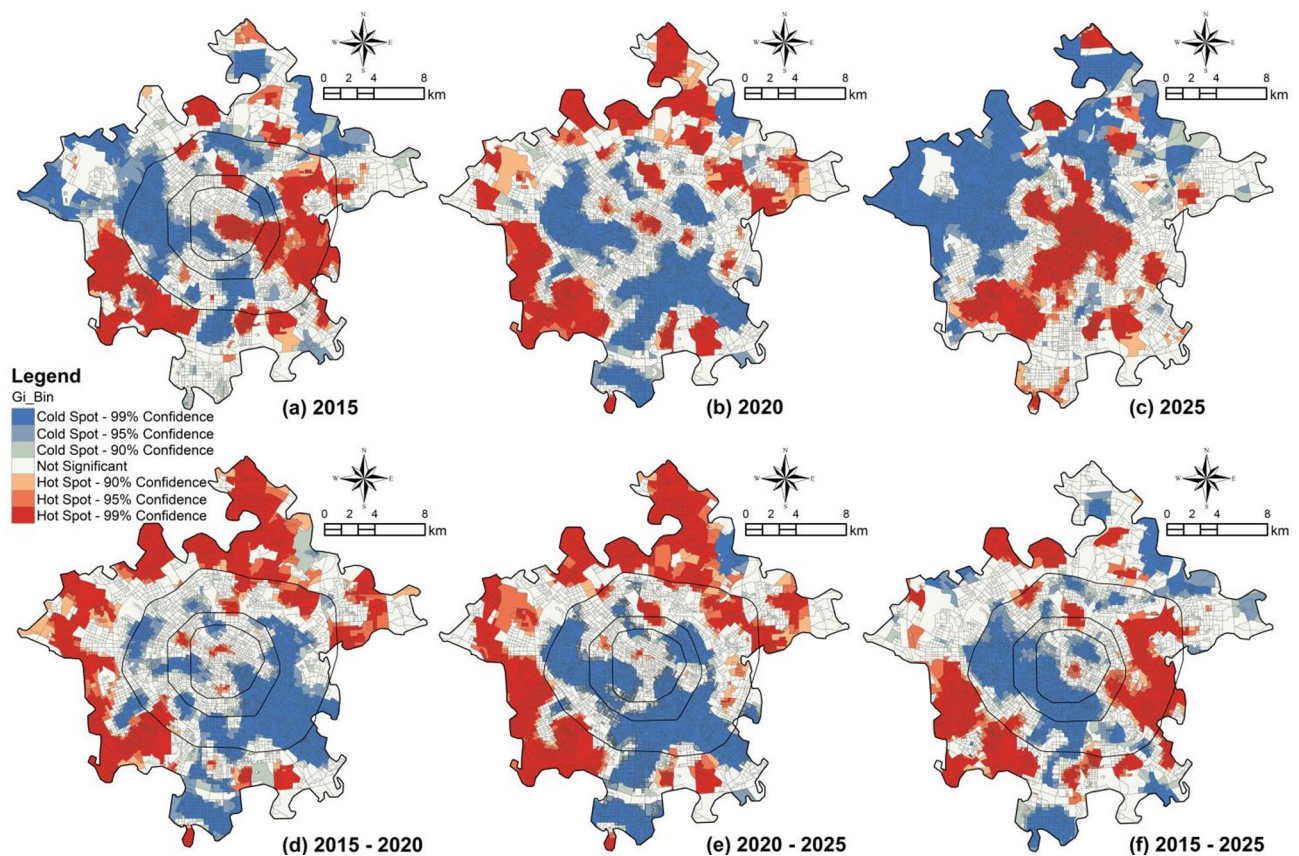


Fig. 4. The spatial distribution of cold and hotspot areas for LST and its rate of change in Chengdu: (a–c) hot and cold spots distribution of static LST in 2015, 2020 and 2025, respectively; (d–f) hot and cold spots distribution of LST change rates during 2015–2020, 2020–2025 and 2015–2025, respectively.

and 17.92% in the three periods, respectively. Although the average value in 2020 was slightly lower than in 2015, the number of blocks with high impervious surface coverage increased, especially in the city center and surrounding blocks. This trend intensified further in 2025, highlighting ongoing construction activities in urban fringe and continued urban sprawl. The proportion of water bodies within blocks was generally low across the three periods, with average values of 0.005%, 0.004% and 0.006%, respectively. These water bodies are mainly concentrated in blocks containing natural water systems. Compared to 2015, the proportions of water body decreased in 2020, with the maximum block level proportion dropping from 4.68% to 1.49%. By 2025, a slight increase in water body coverage was observed relative to 2020, with the maximum block proportion rising to 2.13%. This partial recovery was primarily concentrated in key ecological zones in Jinjiang district and Chenghua district. Regarding the proportion of forest and grassland area, the average proportions at the block-level were 0.0065%, 0.0007% and 0.0637%, respectively, with maximum values of 10.93%, 0.73% and 10.93% across the three periods, respectively. Overall, the forest and grassland coverage showed a “V-shaped” fluctuation trend, with vegetation coverage remaining low in most blocks. High-coverage blocks were primarily located in ecological protection zones and large parks on the urban periphery. Compared to 2015, a decrease in forest and grassland areas was observed in 2020, as urban development encroached upon peripheral green spaces, while ecological spaces within the urban core area were largely preserved. By 2025, a substantial increase in forest and grassland coverage occurred, driven by ecological restoration efforts that expanded green area in several peripheral blocks.

(2) Building morphology factors.

Figure 6 presents the spatiotemporal variation characteristics in various building morphology factors across blocks in Chengdu for the years 2015, 2020 and 2025. Regarding building height, the average values during the three periods were 31.22 m, 29.27 m and 35.17 m, respectively, exhibiting a trend of decline followed by an increase. Spatially, changes in building height displayed a pattern of “central breakthrough and peripheral stability” pattern. In 2015, building heights in the core area ranged from 45.08 m to 81 m, which increased to a range of 58.56 m to 96.21 m by 2025, with certain localized areas characterized by super high-rise buildings exceeding 96 m. The Tianfu Square–Chunxi Road business district and the Financial City area, functioning as dual-core growth hubs, experienced increases in building height of up to 67%. In the eastern part of the city beyond the Second Ring Road, high-rise buildings were distributed in an arc-shaped pattern along major

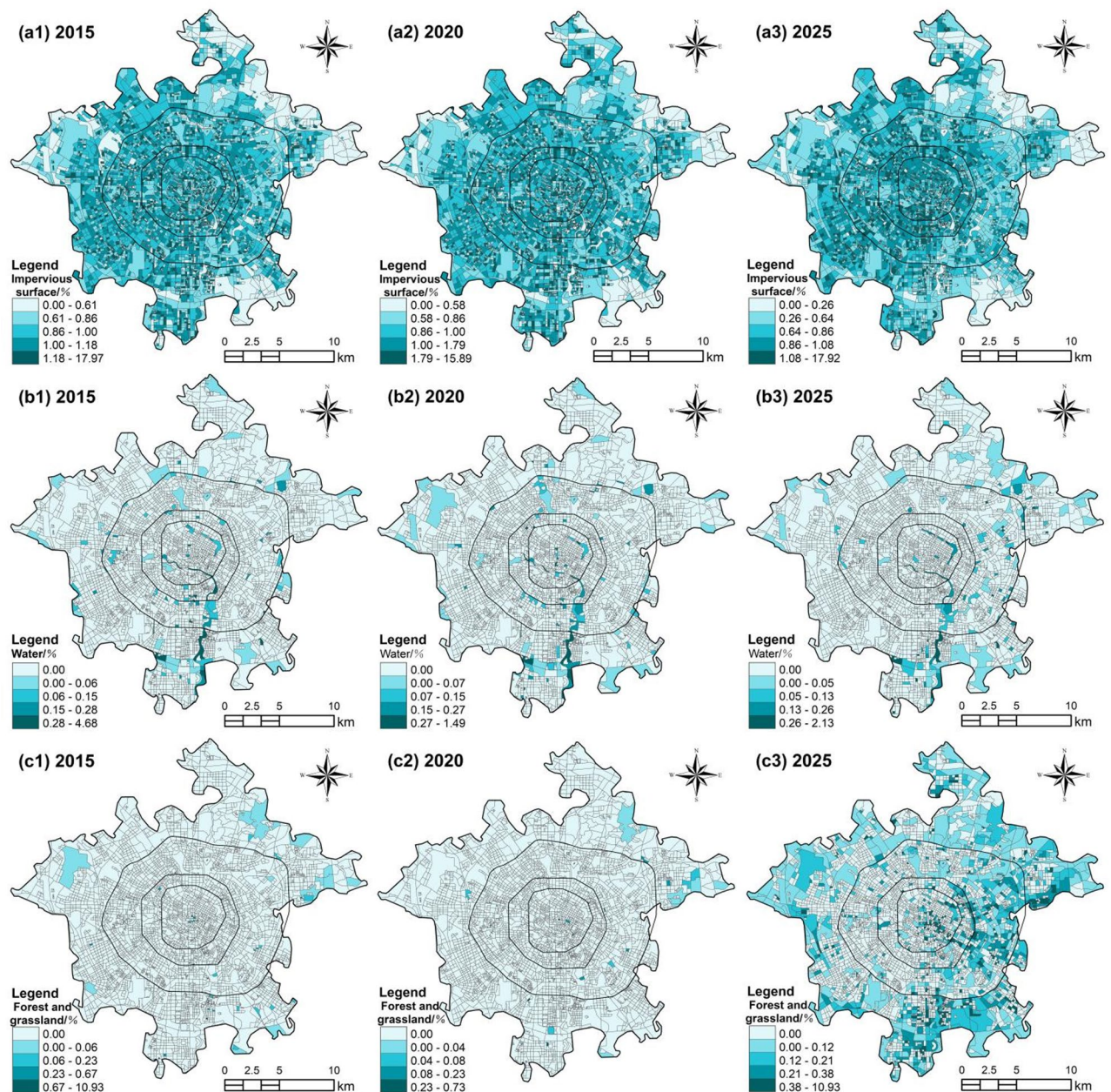


Fig. 5. The spatiotemporal distribution of land cover factors in Chengdu: **(a1–a3)** Proportion of impervious surface area in 2015, 2020 and 2025, respectively; **(b1–b3)** Proportion of waterbody area in 2015, 2020 and 2025, respectively; **(c1–c3)** Proportion of forest and grassland area in 2015, 2020 and 2025, respectively.

ivers, while most areas within the Third Ring Road maintained low-rise buildings below 15 m. Concerning building density, the average density for the three periods was 0.210%, 0.183% and 0.196%, respectively, showing a general decreasing trend. Spatially, building density exhibited a significant pattern of “dense center and sparse periphery”. High-density blocks were mainly concentrated within the First and Second Ring Roads, with a “higher in the west, lower in the east” distribution within the Third Ring Road. This variation was largely influenced by differences in urban functional layout and land use. In 2015, building density in the core area ranged from 47% to 93%, improving to 45%–87% by 2025. The decrease in building density in central urban districts coincided with an increase in building height, indicating a trend toward vertical intensification of urban space. Peripheral urban areas maintained relatively low building densities, reflecting effective regulation of floor area ratio to preserve open urban space. This spatial differentiation represents the synergistic effect of compact urban core space utilization and peripheral landscape management under the city’s “medium priority” development strategy. The average SVF values for the three periods were 0.0766, 0.091 and 0.081, respectively, showing a spatial pattern of gradual increase from the urban core toward the periphery. The urban core areas

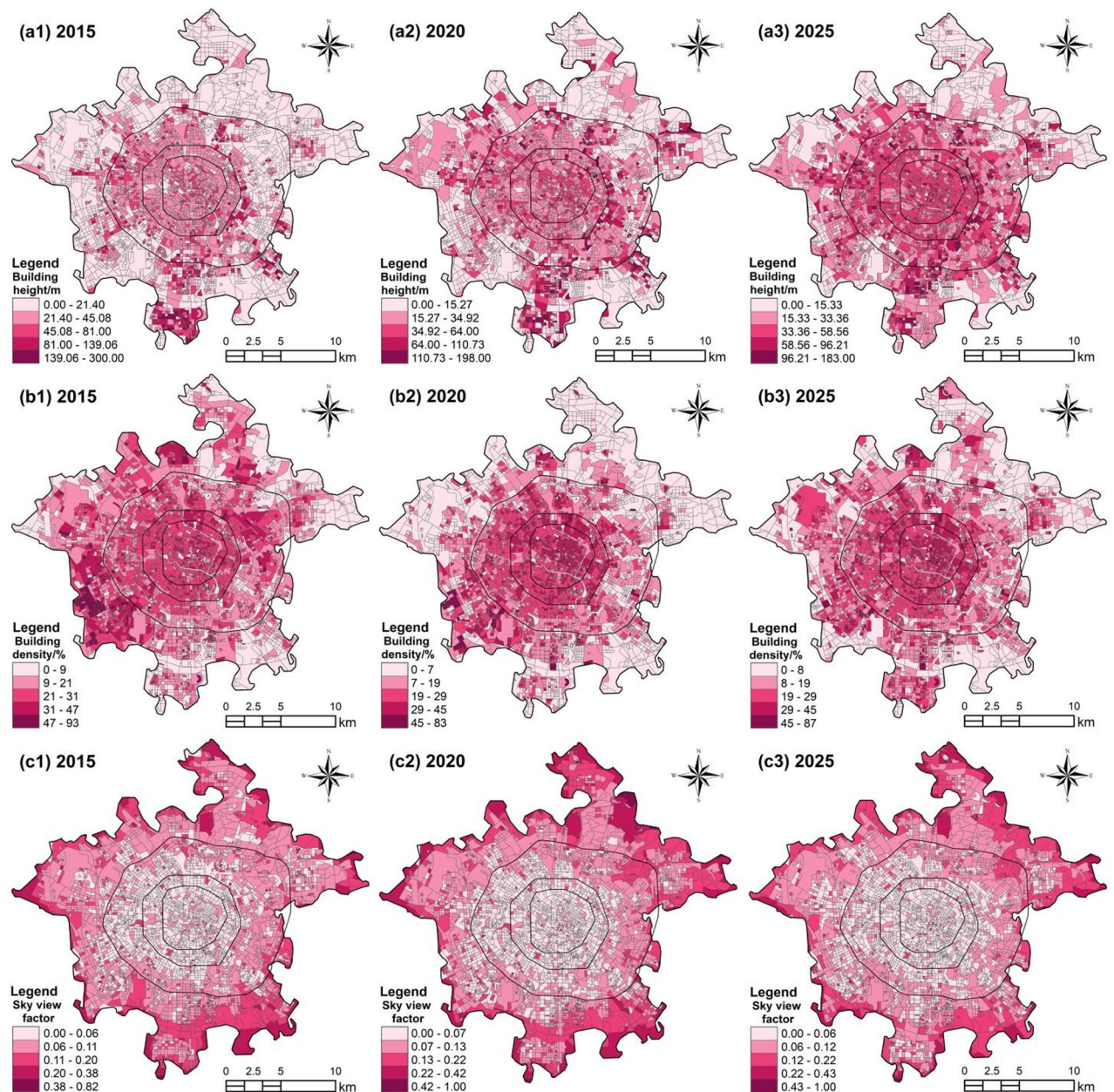


Fig. 6. The spatiotemporal distribution of building morphology factors in Chengdu: **(a1–a3)** Average building height in 2015, 2020 and 2025, respectively; **(b1–b3)** Building density in 2015, 2020 and 2025, respectively; **(c1–c3)** Sky view factor in 2015, 2020 and 2025, respectively.

exhibited low SVF values ranging from 0.00 to 0.11, indicating a significant urban canyon effect resulting from high-density construction, which significantly limits sky visibility.

(3) Road traffic factors.

Figure 7 illustrates the spatiotemporal variation characteristics of road density across the block of Chengdu in 2015, 2020 and 2025. Quantitatively, the average road density increased from 0.008 km/km^{-2} in 2015 to 0.024 km/km^{-2} in 2020 and further to 0.032 km/km^{-2} in 2025. The maximum value rose from 6.77 km/km^{-2} in 2015 to 9.03 km/km^{-2} in 2025, corresponding to an average annual growth rate of 3.2%. This growth trend indicates the ongoing expansion and improvements of Chengdu's road network and transportation infrastructure. In terms of spatial pattern evolution, road density generally exhibits a pattern of central intensification and peripheral expansion. Core areas such as Jinjiang, Qingyang and Wuhou districts showed sustained increases in road density, maintaining relatively high levels. Additionally, the southern part of the study area, centered on the Tianfu Avenue development corridor, experienced significant increases in surrounding districts. Although

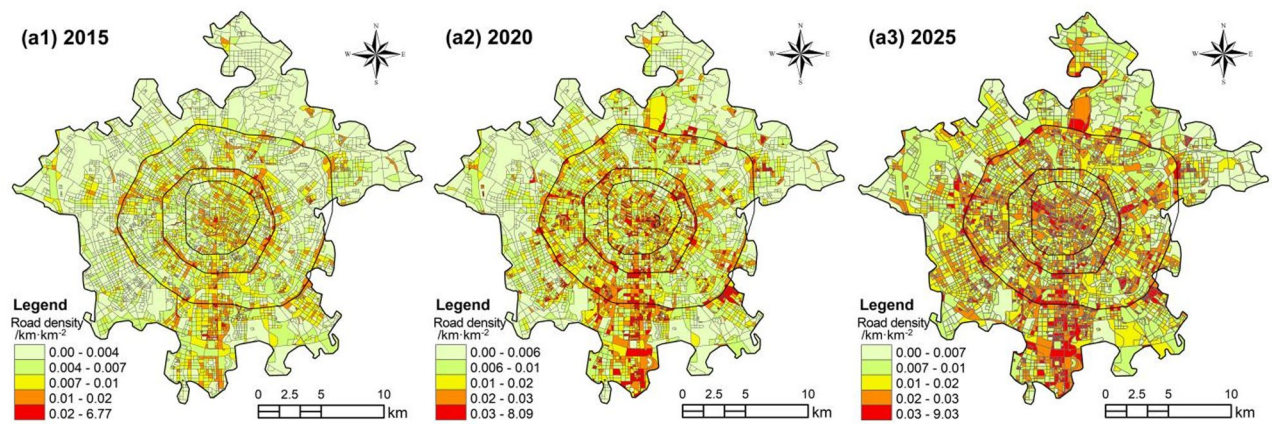


Fig. 7. The spatiotemporal distribution of road traffic factors in Chengdu: (a1–a3) road density in 2015, 2020 and 2025, respectively.

road density within the Third Ring Road remains relatively low, recent years have seen improvements through new road construction and systematic upgrades of existing roads, gradually enhancing connectivity and the overall transportation network. Overall, the growth and spatial distribution of road density reflect Chengdu's coordinated efforts under the “Central Priority” development strategy to optimize the transportation network while reshaping the urban spatial structure. On one hand, the road system optimization in central urban areas enhances spatial efficiency; on the other hand, the extension of road networks in peripheral areas supports the expansion of the urban scale.

Characteristics of changes in comprehensive UPF

Figure 8 presents the spatial distribution pattern of comprehensive UPF for the year 2015, 2020 and 2025 along with their corresponding change rates. Quantitatively, the average UPF values were 0.084, 0.098 and 0.108 for the three periods, respectively, indicating a continuous intensification of urban construction that line with the city's ongoing development. Spatially, the UPF exhibits a distinct pattern of high values in the center and low values in the periphery. Although the overall spatial patterns of UPF and LST differ, certain localized similarities can still be observed. At single time points, the high value UPF blocks in 2015 were scattered in a point-like distribution, largely overlapping with areas of high LST. Blocks with low UPF values were mainly located in peripheral areas such as the Tianhui Town and the Fenghuangshan in Jinniu district, as well as the Longtan and Bailianchi in Chenghua district. Compared to 2015, the spatial extent of high UPF value blocks expanded by 2020, while numerous low-value districts emerged in the urban periphery, such as the Wenjia in Qingyang district, indicating an improvement in the optimization of UPF in peripheral areas. By 2025, the spatial distribution of UPF remained broadly consistent with 2020, continuing decreasing gradient from the urban core to the outskirts.

Across the three periods (2015–2020, 2020–2025 and 2015–2025), the change rates of UPF varied considerably across blocks, with average values of 0.587, 0.384 and 0.694, respectively. Generally, blocks with greater changes in UPF were located in peripheral areas with lower construction intensity and certain ecological spaces, whereas central districts with higher construction intensity showed relatively smaller changes. From 2015 to 2020, blocks with both high and low UPF change rates were fairly evenly distributed across the study area, with high-value blocks mostly found at the urban periphery. From 2020 to 2025, the UPF changes in the urban core tended to stabilize, while peripheral blocks experienced more significant changes, especially in low-density construction zones such as the Tianhui Town block in Jinniu district, the Wenjia block in Qingyang district and the Longtan block in Chenghua district. Over the long term, from 2015 to 2025, blocks with larger UPF changes were concentrated in newly developed peripheral areas.

The global spatial autocorrelation results for comprehensive UPF and its rate of change (Table 4) show that the Moran's I values for both static and dynamic measurements are positive, with Z -values exceeding 2.58 and passing the significance test at $P < 0.01$. This demonstrates a significant positive spatial autocorrelation of comprehensive UPF and its rate of change, meaning that blocks with high or low construction intensity and its changes tend to be spatially adjacent and clustered. Regarding changes in comprehensive UPF, the Moran's I value showed an upward trend from 2015 to 2020, indicating a rise in spatial clustering during this period. However, from 2020 to 2025, the Moran's I value declined, indicating a decline in spatial clustering. For the change rates, the Moran's I value peaked during 2015–2020, followed by 2020–2025 and 2015–2025, indicating the strongest spatial clustering of UPF change rates in the earlier period. Comparatively, the Moran's I values of comprehensive UPF and its change rates were consistently lower than those of LST and its changes for the corresponding periods, indicating that the low and high values of LST and its evolution show stronger spatial clustering than those of comprehensive UPF and its evolution.

Figure 9 illustrates the spatial distribution of hot and cold spots for the comprehensive UPF and its rate of change. At single time points, the hot and cold spot blocks across all three years exhibit distinct concentric ring patterns. Hot spot blocks are concentrated in the urban core areas, characterized by high proportions of

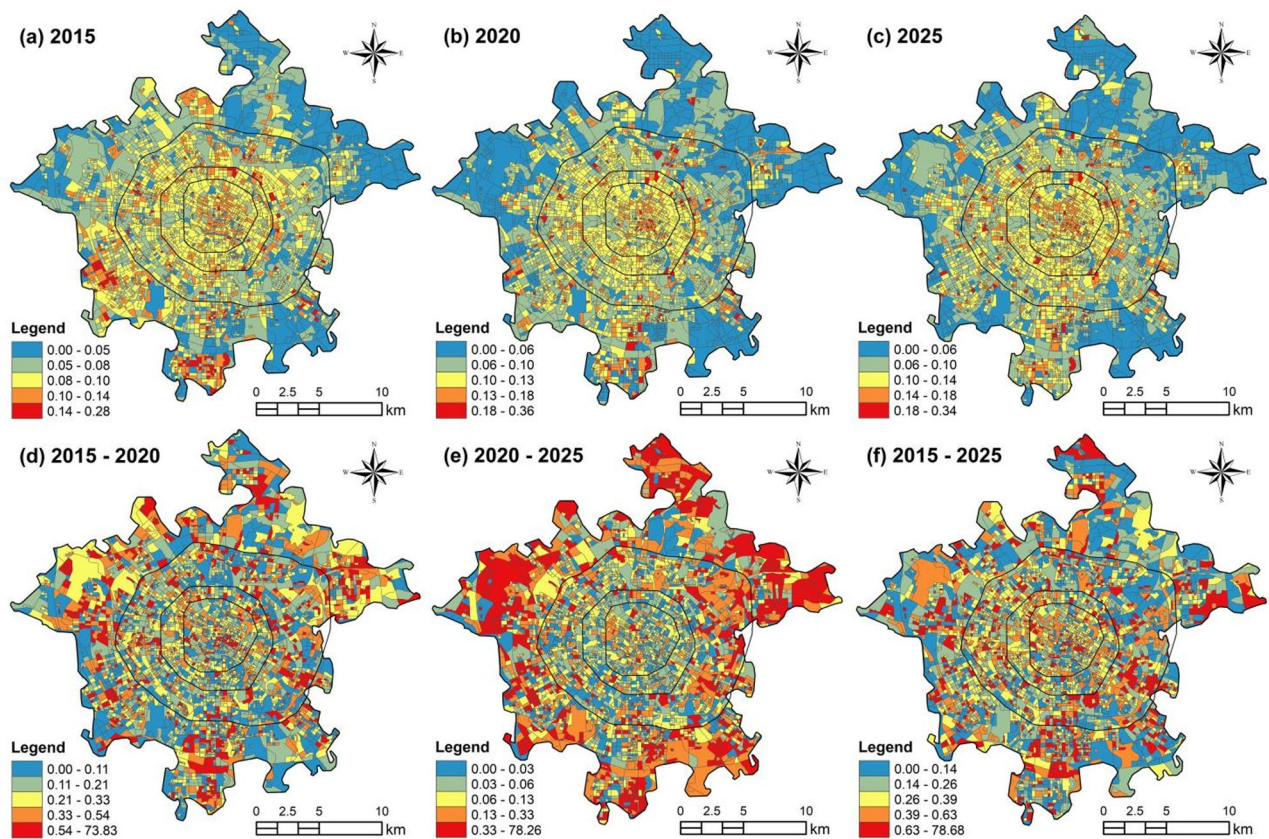


Fig. 8. The spatiotemporal distribution of comprehensive physical form and its rate of change in Chengdu: (a–c) Represent the spatial distribution of static Comprehensive UPF for the years 2015, 2020 and 2025, respectively; (d–f) show the spatial distribution of Comprehensive UPF change rates for the periods 2015–2020, 2020–2025 and 2015–2025, respectively.

	Year	Global Moran's I	Z value	P value
Urban physical form	2015	0.195 8	53.189 4	0.000 0
	2020	0.247 9	67.339 8	0.000 0
	2025	0.210 6	57.216 9	0.000 0
	2015–2020	0.034 8	9.839 6	0.000 0
	2020–2025	0.031 9	9.643 0	0.000 0
	2015–2025	0.017 7	5.037 1	0.000 0

Table 4. Analysis results of global spatial autocorrelation.

impervious surfaces, dense buildings, greater average building heights and dense road networks. In contrast, cold spots are mainly located in the urban outskirts, where impervious surface proportions are lower, building heights rarely exceed 20 m, building density is typically below 9% and the SVF values are relatively high. Regarding the spatial pattern of change rate, most blocks across the three periods exhibited no significant spatial clustering, with hot spots limited to a few localized blocks and cold spots scattered without clear concentration. This indicates that the spatial distribution of UPF changes is generally random or dispersed, reflecting a lack of coordinated urban planning strategies to guide the evolution of UPF.

Coupling and coordination characteristics between urban LST and comprehensive UPF
Coupling degree

(1) Static coupling degree.

The static coupling degree between the LST and UPF exhibited an upward trend (Fig. 10). The average coupling degrees of blocks in 2015, 2020 and 2025 were 0.3689, 0.3810 and 0.3912, respectively, indicating an increasing spatial influence of UPF on the thermal. Regarding changes in coupling types, the transition mainly occurs

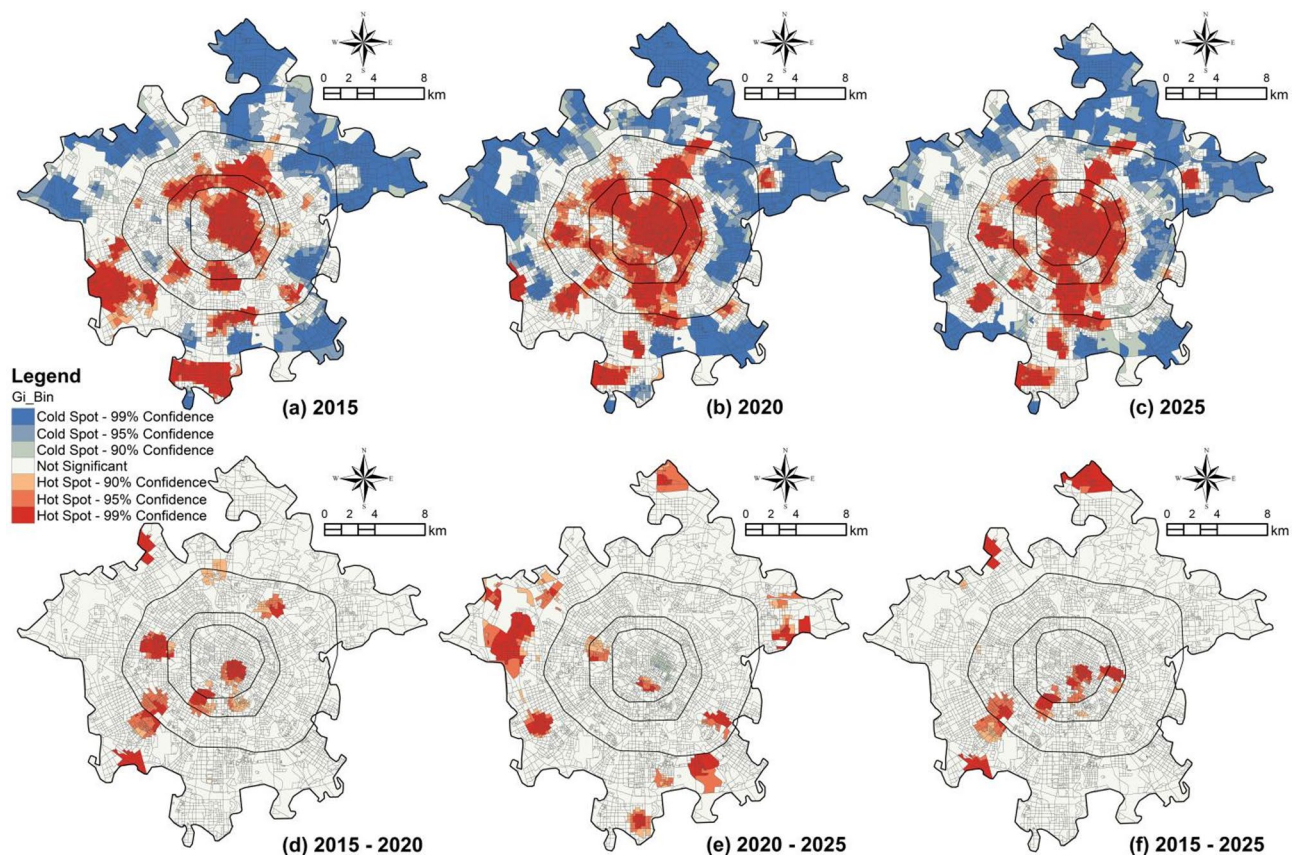


Fig. 9. The spatial distribution of cold and hotspot areas for comprehensive physical form and its rate of change in Chengdu: (a–c) Hot and cold spots distribution of static Comprehensive UPF in 2015, 2020 and 2025, respectively; (d–f) Hot and cold spots distribution of Comprehensive UPF change rates during 2015–2020, 2020–2025 and 2015–2025, respectively.

between the high-level coupling stage and the adaptation stage, with the adaptation stage being dominant overall. In 2015, 3851 blocks accounting for 75% of the total 5135 blocks, were in the adaptation stage. The number of blocks with low coupling degree was the lowest, showing a decreasing trend from 57 in 2015 to 29 in 2020 and further down to 13 in 2025. From a spatial perspective, a clear pattern emerged: the urban center is characterized by a high-level coupling stage, the near-central districts were mainly in the adaptation stage and the outer suburbs were mostly in the antagonistic stage. Due to the high construction intensity and limited ecological spaces in the urban center, the overall LST remains relatively high, leading to a strong coupling relationship with the UPF. In contrast, the outer suburbs, despite having lower UPF indices, still experience relatively high LST, resulting in a predominance of the antagonistic stage. In 2015, high-level coupling areas were mainly concentrated in Wuhou district and several core urban zones. By 2020, the spatial extent of these areas expanded, with small-scale clustering observed within the city center. By 2025, the coverage of high-level coupling areas continued to expand, indicating a continued spatial intensification of the coupling relationship between LST and UPF.

(2) Dynamic coupling degree.

The dynamic coupling degree between the LST and UPF varies considerably across different periods. The average dynamic coupling degrees for the periods 2015–2020, 2020–2025 and 2015–2025 were 0.49, 0.34 and 0.38, respectively. From 2015 to 2020, the coupling predominantly remained in the adaptation stage, with 2599 blocks (51% of the total) classified in this category. In contrast, from 2020 to 2025, the number of blocks with low-level coupling significantly increased and became dominant, with 2,498 blocks (48.64% of the total). This shift is mainly due to Chengdu's promotion of UPF evolution through spatial structures optimization and strengthened land use policies during this period, while the response of LST slightly lagged behind. Consequently, the dynamic coupling level between the two systems decreased. Some peripheral blocks exhibited elevated dynamic coupling degrees, such as the eastern edge of Chenghua district, the northern part of Jinniu district and the border areas between Qingyang and Wuhou districts. From 2015 to 2025, the antagonistic stage was dominant, with 3,030 blocks accounting for 59%. Low-level coupling mainly occurred in the northeastern peripheral area. Notably, certain blocks with relatively high static coupling degrees did not show correspondingly high dynamic coupling, indicating that static spatial synergy does not necessarily imply temporal synchrony. In summary, the dynamic

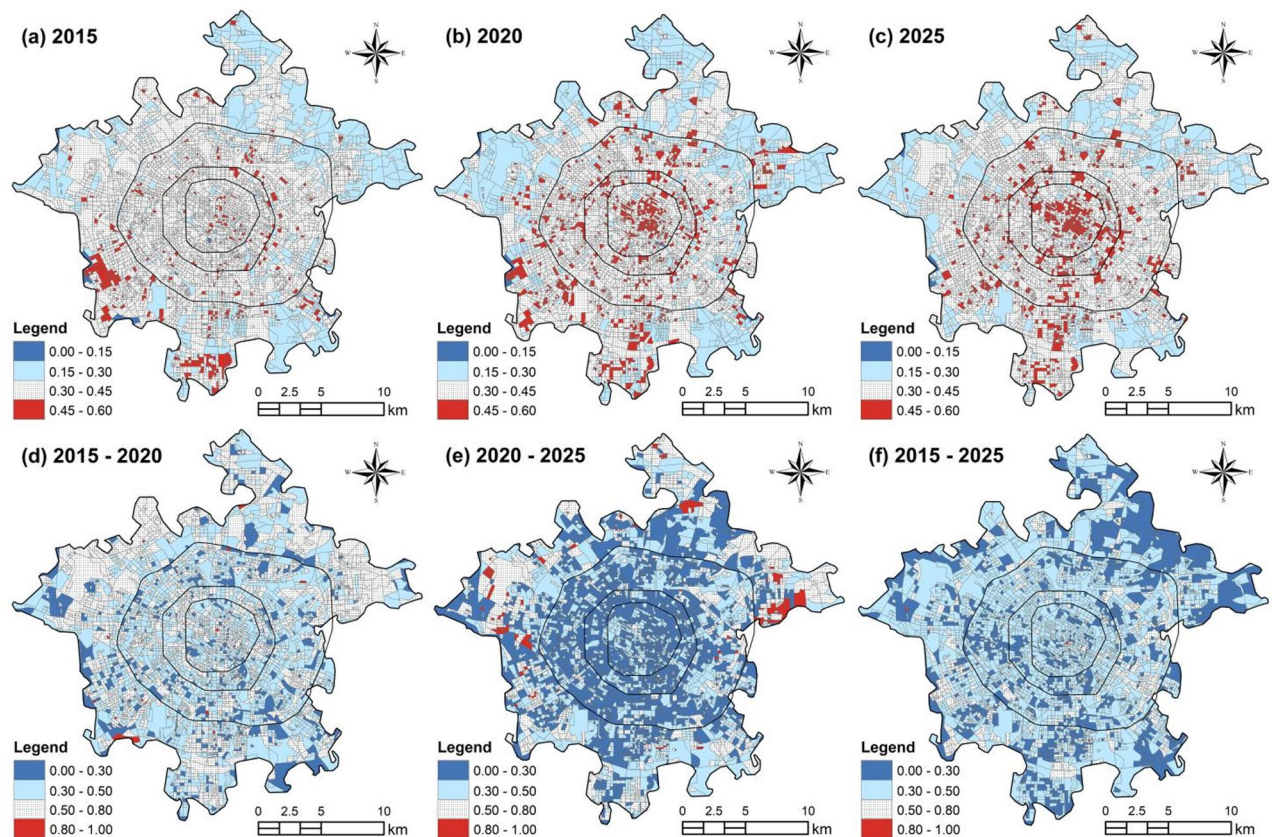


Fig. 10. The coupling degree between LST and comprehensive UPF in Chengdu: (a–c) Static coupling degree for 2015, 2020 and 2025, respectively; (d–f) Dynamic coupling degree for 2015–2020, 2020–2025 and 2015–2025, respectively.

coupling between LST and UPF remains relatively low and shows a declining trend. This asynchrony suggests that, despite significant improvements in urban spatial form through morphological optimization, the thermal environment response lags behind, reflecting a delayed adaptation of LST to spatial restructuring.

Coupling coordination degree

• (1) Static coupling coordination degree.

The static coupling coordination degree between the LST and UPF shows a declining trend (Fig. 11). The average value for blocks in 2015, 2020 and 2025 were 0.3377, 0.3207 and 0.3021, respectively. Analyzing these trends alongside the variations in LST and UPF reveals that the decline from 2015 to 2020 reflects a negative outcome, indicating that changes in UPF during this period failed to effectively improve the regulation of the LST. Conversely, the decline from 2020 to 2025 reflects a positive trend, implying that the changes in UPF did not further aggravate the overall urban LST. Spatially, the urban core was predominantly in the adaptation stage, which gradually declined over time. The number of blocks in the coordination stage decreased from 4009 (78.07%) in 2015 to 3,165 (61.61%) in 2020 and further to 2612 (50.86%) in 2025. The urban periphery is dominated by the antagonistic stage, with an increasing trend. The number of blocks in this stage increased from 995 (19.37%) in 2015 to 1776 (34.59%) in 2020 and further to 2345 (45.67%) in 2025. High-level coupling coordination areas are mainly distributed in the urban periphery. In 2015, these areas were concentrated in the Jinhuaqiao block of Wuhou district, where the LST and UPF exhibited a strong coupling interaction. By 2020, the spatial extent of high-level coupling coordination areas expanded and showed certain spatial continuity and stable evolution patterns in areas like the Wukuaishi block in Jinniu district and the Cuiqiao block in Wuhou district. However, by 2025, the spatial coverage of high-level coupling coordination areas shrank, while low-level coupling coordination areas increased, mainly in blocks such as Wenjia in the western Qingyang district and the Xihua in the northern part of Jinniu district.

(2) Dynamic coupling coordination degree.

The dynamic coupling coordination relationship between the LST and UPF exhibits significant stage-specific characteristics. The average value for the periods 2015–2020, 2020–2025 and 2015–2025 were 0.074, 0.054 and 0.382, respectively. These value indicate a lack of short-term coordination but an overall improvement over the

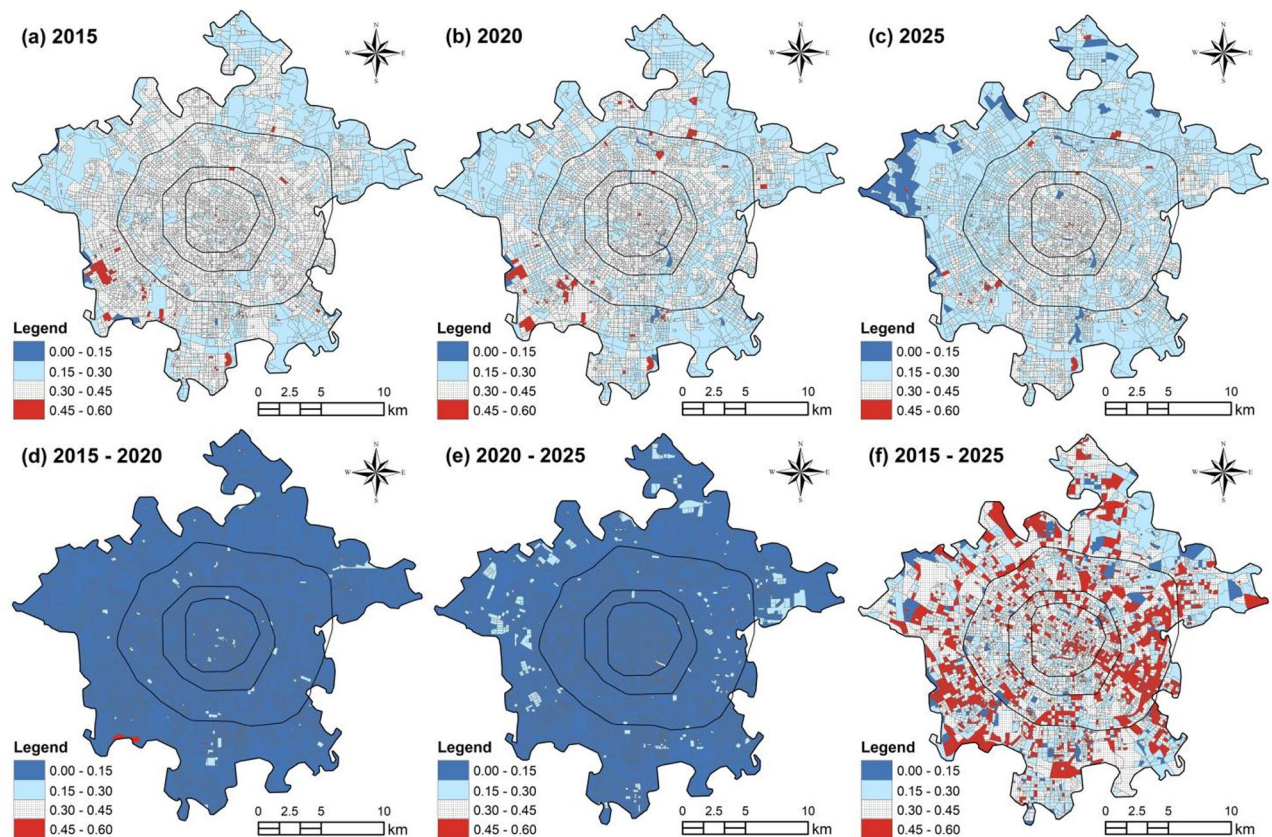


Fig. 11. The coupling coordination degree between LST and comprehensive UPF in Chengdu: (a–c) Static Coupling coordination degree for 2015, 2020 and 2025, respectively; (d–f) Dynamic coupling coordination degree for 2015–2020, 2020–2025 and 2015–2025, respectively.

long term. From 2015 to 2020, most blocks were at a low level of dynamic coupling coordination, with 4988 blocks accounting for 97.14%, far exceeding the number of blocks with low-level static coupling coordination during the same period. From 2020 to 2025, the number of low-level coupling coordination blocks decreased to 4852 (94.98%), while the number of blocks in the antagonistic stage increased to 264 (5.14%). These antagonistic blocks were mainly concentrated in areas experiencing intensive urban renewal and redevelopment, such as the Cuqiao block in Wuhou district, the Wenjia block in Qingyang district and the Tianhuizhen block in Jinniu district. From a long-term perspective (2015–2025), the number of blocks exhibiting high-level dynamic coupling coordination increased significantly to 1380, accounting for 26.87%, with these blocks mainly distributed within areas enclosed by the Third Ring Road. In summary, the coupling coordination degree between changes in the LST and UPF remains generally low across different time periods, especially in peripheral blocks. This phenomenon is mainly due to the continuous optimization of UPF while the improvement in LST has not been as significant, weakening their synergistic evolution. This reflects that the current optimization of urban spatial structure has not yet to effectively influence the regulation of the thermal environment. It highlights the urgent need for developing adaptive and responsive spatial planning mechanisms to improve the dynamic coordination between urban morphology and thermal performance.

Discussion

Advantages of this study

This study focuses on the spatiotemporal patterns of urban LST and UPF, alongside their static and dynamic coupling coordination relationships. Compared with previous studies, it offers several key advantages. First, it integrates multi-source spatial data, enhancing the analytical capability for evaluating both the urban LST and UPF. Specifically, this study synthesizes remote sensing imagery, land use data, building morphology and road traffic information. By retrieving LST data and extracting urban morphological indicators, this study achieves high-precision mapping and quantitative characterization of the spatiotemporal patterns of both LST and UPF. While earlier studies have primarily focused on the spatiotemporal evolution of LST⁵⁰, this research highlights the dynamic impact of UPF and its changes on LST. Secondly, this study constructs a multi-temporal coupling coordination degree model to explore the spatiotemporal dynamic coupling coordination mechanism. While most previous studies have focused on spatial pattern analysis at a specific time point or on dynamic decoupling within a single time period⁵¹, this study analyzed data from three time points—2015, 2020 and 2025—to identify the spatiotemporal dynamic evolution characteristics of LST and UPF. Based on this, both coupling

degree and coupling coordination degree models are applied to systematically assess their static alignment and dynamic coupling coordination. This approach overcomes traditional methods that mainly rely on correlation coefficients⁵² or parameter tests⁵³ and provides new insights into understanding the synergistic development of urban thermal environments and UPF. Finally, this study employs the block as the fundamental unit of analysis to enhance the practical applicability of the research outcomes for urban planning practices. Traditional research units based on administrative divisions or regular grids often overlook the LST differences between individual blocks⁵⁴. In contrast, using block units allows for a more precise identification of the spatial heterogeneity in the coupling coordination between the LST and UPF, offering valuable spatial insights to inform urban planning decisions.

Implications for future urban planning

Firstly, in terms of the static spatial distribution characteristics of urban LST and UPF, a general spatial mismatch is observed in peripheral urban blocks, where the physical form is underdeveloped, but the LST remains high. This results in low static coupling and static coupling coordination degree in these areas. Secondly, with regard to dynamic changes, the comprehensive UPF in the peripheral blocks shows a significant decreasing trend, while the LST exhibits a lagged response, highlighting a disconnect between spatial form optimization and thermal environmental adaptation. Lastly, the dynamic coupling coordination degree between urban LST and UPF displays the pattern of “small short-term fluctuations and significant long-term variations” indicating a lack of stable and sustained linkage between thermal environment governance and spatial evolution. Based on these findings, future urban planning should enhance response mechanisms from the following strategies: (1) Optimize the spatial structure in the urban peripheral areas. Blocks with low physical form levels and elevated heat risk—such as Tianhuizhen, Xihua, Jinquan, Huangtanba, Wenjia, and Longtan—should be included into the regulatory scope for the city’s land use structure. By systematically integrating ventilation corridors, green space systems, permeable pavement and other forms of green infrastructure, the capacity for thermal environment regulation and ecological service provision can be effectively strengthened. (2) Enhance the timeliness of UPF adjustments by establishing a dynamic regulation framework that incorporates real-time feedback from thermal environment. This approach will improve the responsiveness and adaptability of the urban thermal environment. For example, within Chengdu’s Fourth Ring Road, urban planning should proactively address thermal environment changes through continuous thermal monitoring, adaptive zoning strategies and sustainable construction practices that prioritize both environmental resilience and urban livability. (3) Establish long-term linkage mechanisms between spatial structure evolution and thermal environment improvement. Urban renewal and development of green infrastructure⁵⁵ should be promoted, using blocks as the fundamental planning units to facilitate the coordinated evolution of spatial form and urban climate. In the long term, an integrated urban spatial governance framework should be established, with a focus on achieving coupling coordination between UPF optimization and thermal environment regulation. This will support the city’s transition climate resilience and spatial sustainability.

Limitations and future prospects

Although this study has made progress in revealing the coupling and coordination mechanisms between the urban LST and UPF, several limitations remain: Firstly, the time span of the study is relatively limited, with the analysis primarily based on remote sensing imagery and built environment data from 2015, 2020 and 2025. While this provides insight into the general evolution trends of LST and UPF, the quality of the remote sensing imagery limits our ability to capture more detailed inter-annual variations or seasonal fluctuations. Additionally, the building information used in this study is sourced from Gaode Maps. However, due to the absence of historical building data, it was not possible to accurately obtain building information from earlier years. As a result, we were limited to analyzing changes in UPF based on data from only these three time points, which may restrict the comprehensiveness of the long-term change analysis. Secondly, this study used the 2022 land cover dataset provided by Global Land Cover to replace the 2025 data for analysis. This substitution was made because the 2022 dataset closely mirrors the 2025 data in terms of spatial characteristics and land cover types. While the analysis results for all land cover factors generally align with the actual trends, it is important to note that some dynamic changes might have been overlooked due to the use of proxy data. Thirdly, this study focuses mainly on the impact of UPF elements on the urban thermal environment, without fully accounting for anthropogenic factors such as population density, traffic emissions and energy consumption. These human-driven factors also play a significant role in shaping and evolving urban thermal environments, and their inclusion would provide a more comprehensive understanding of the system dynamics. In light of these limitations, future research could be expanded in the following directions: (1) Integrate multi-temporal remote sensing data with higher spatiotemporal resolution and extend the time span of analysis to construct more detailed evolution sequences of the LST and UPF. This would enhance the ability to detect and analyze dynamic coupling changes over a longer period. (2) Incorporating socioeconomic and anthropogenic activity indicators into the analytical framework to more comprehensively reveal the coupled effects of natural and human processes in shaping the urban thermal environment.

Conclusions

This study focuses on the blocks within the five central districts of Chengdu, analyzing the spatiotemporal dynamic evolution characteristics of LST and UPF, as well as their dynamic-static coupling coordination relationship. The following conclusions are drawn:

- (1) The evolution trends of LST and UPF exhibit spatiotemporal differences. LST follows a “rise-then-fall” trend, with its spatial pattern shifting from “center-low, edge-high” to “center-high, edge-low”. In both 2015

and 2020, LST hotspot blocks were mainly distributed in fragmented patches at the urban periphery, while by 2025, these hotspots coalesced into a continuous patch concentrated in the urban core. Cold spot blocks gradually showed continuous, patch-like features in the urban periphery. Overall, UPF showed a steady upward trend, forming a “center-high, edge-low” ring structure. This pattern highly correlated with the spatial distribution of both LST hotspots and cold spots.

- (2) There is spatial consistency between the changes in LST and UPF, both exhibited a “central stability, edge fluctuation” pattern. In the short-term phase, LST change hotspots and cold spots showed a continuous patch-like distribution, while in the long-term phase, these patterns became more dispersed. The spatial aggregation of UPF changes was less pronounced, with only a few localized blocks showing hotspots, and no notable aggregation of cold spots.
- (3) The static coupling degree between LST and UPF showed an overall upward trend. Spatially, the central urban area primarily fell into the high-level coupling phase, while near-central areas were in the adaptation phase and suburban areas were an antagonistic phase. Dynamic coupling degrees varied significantly across different time phases: from 2015 to 2020, it was primarily in the adaptation phase; from 2020 to 2025, it shifted to a dominated of low-level coupling; and from 2015 to 2025, the long-term evolution was mainly in the antagonistic phase.
- (4) The static coupling coordination degree between LST and UPF showed a downward trend. The city center was predominantly in the adaptation phase, while the periphery was mainly in the antagonistic phase. High-level coupling coordination areas were primarily distributed in point-like patterns at the urban periphery, gradually expanding outward. The dynamic coupling coordination degree was generally lower than the static one, dominated by low-level coupling coordination and exhibiting the characteristic of “small short-term fluctuations, significant long-term fluctuations”.

Data availability

No, I do not have any research data outside the submitted manuscript file.

Received: 2 July 2025; Accepted: 4 November 2025

Published online: 12 December 2025

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Funding

This study was supported by the National Natural Science Foundation of China (Grant Number: 42501258); Research Center for Human Geography of Tibetan Plateau and Its Eastern Slope, Chengdu University of Technology (Grant Number: RWDL2025-YB001).

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-025-27517-5>.

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